

Practical 7: ARIMA Modeling of Commercial Bank Real Estate Loans

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1. Question

Consider the monthly volume of commercial bank real estate loans (in billions of dollars) data set stored in a text file. Write the R program for the following:

- (f) Import data into the R environment.
 - (g) Convert the data into a time series object.
 - (h) Plot the data to identify the dominant component.
 - (i) Check stationarity or non-stationarity using ACF/PACF plot.
 - (j) Check stationarity or non-stationarity using the Augmented Dickey-Fuller (ADF) test.
 - (k) If data is non-stationary, make it stationary using an appropriate operator.
 - (l) Based on the dominating component, select the suitable model to fit the data and finalize the order of the model using ACF/PACF plots with model selection criteria such as AIC.
 - (m) Fit the data using the selected model and estimate the parameters of the model.
 - (n) Check the goodness of fit of the model.
-

2. R Code (Question-wise Solution)

```
# To delete previous history or stored variables
rm(list = ls(all = TRUE))

library(tseries)
library(forecast)
```

```
# -----
# (f) Import data into the R environment
# -----
```

```
bank_data <- as.ts(scan("bank_case.txt"))
```

```
# Printing the data
bank_data
```

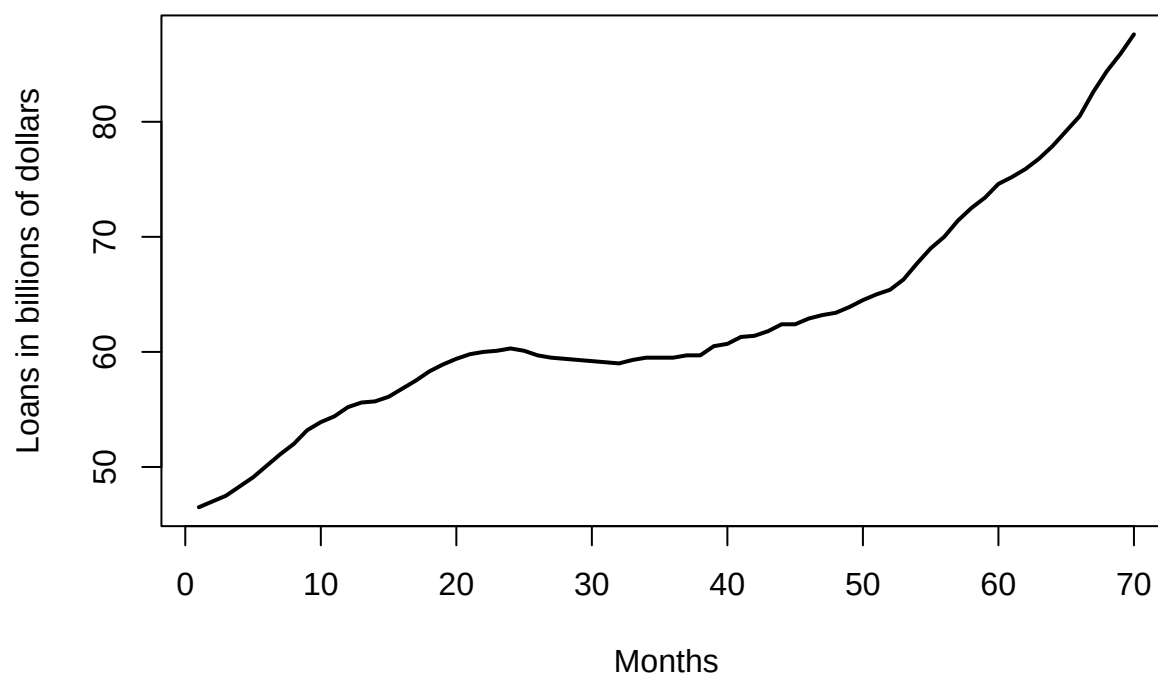
```
## Time Series:
## Start = 1
## End = 70
## Frequency = 1
## [1] 46.5 47.0 47.5 48.3 49.1 50.1 51.1 52.0 53.2 53.9 54.4 55.2 55.6 55.7 56.1
## [16] 56.8 57.5 58.3 58.9 59.4 59.8 60.0 60.1 60.3 60.1 59.7 59.5 59.4 59.3 59.2
## [31] 59.1 59.0 59.3 59.5 59.5 59.5 59.7 59.7 60.5 60.7 61.3 61.4 61.8 62.4 62.4
## [46] 62.9 63.2 63.4 63.9 64.5 65.0 65.4 66.3 67.7 69.0 70.0 71.4 72.5 73.4 74.6
## [61] 75.2 75.9 76.8 77.9 79.2 80.5 82.6 84.4 85.9 87.6
```

```
# -----
# (g) Convert data into time series object
# -----
# (Already converted using as.ts())
```

```
# -----
# (h) Plot the data to identify dominant component
# -----
```

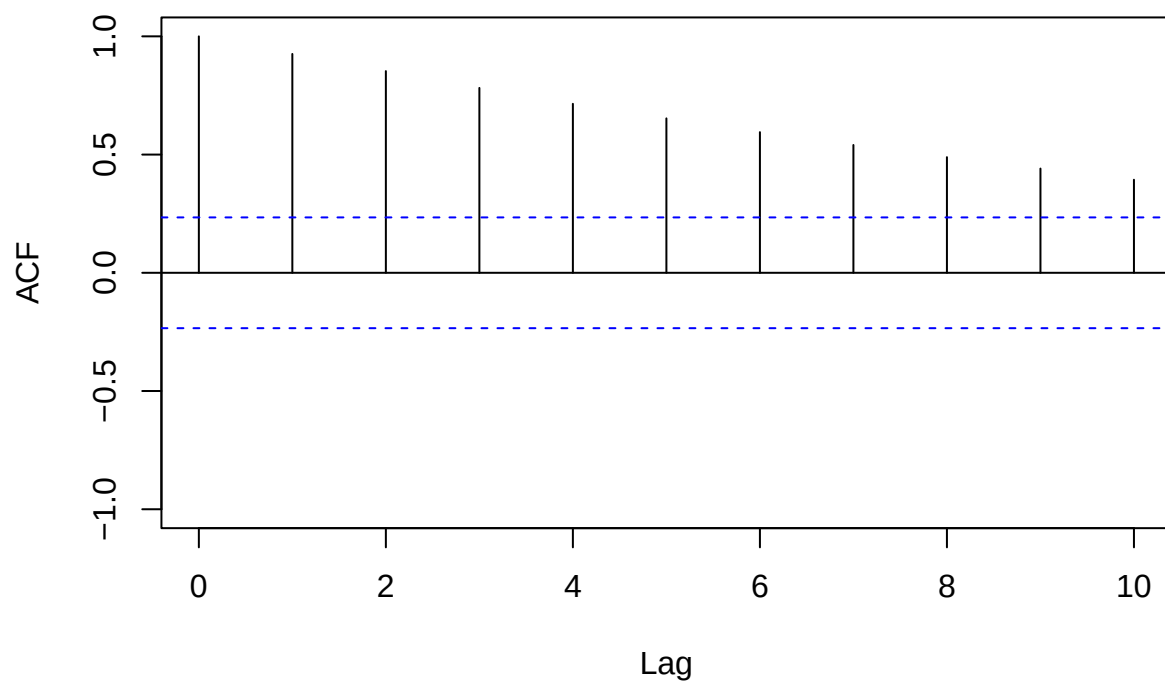
```
plot(bank_data,
     main = "Commercial Bank Real Estate Loan Data",
     xlab = "Months",
     ylab = "Loans in billions of dollars",
     lwd = 2)
```

Commercial Bank Real Estate Loan Data



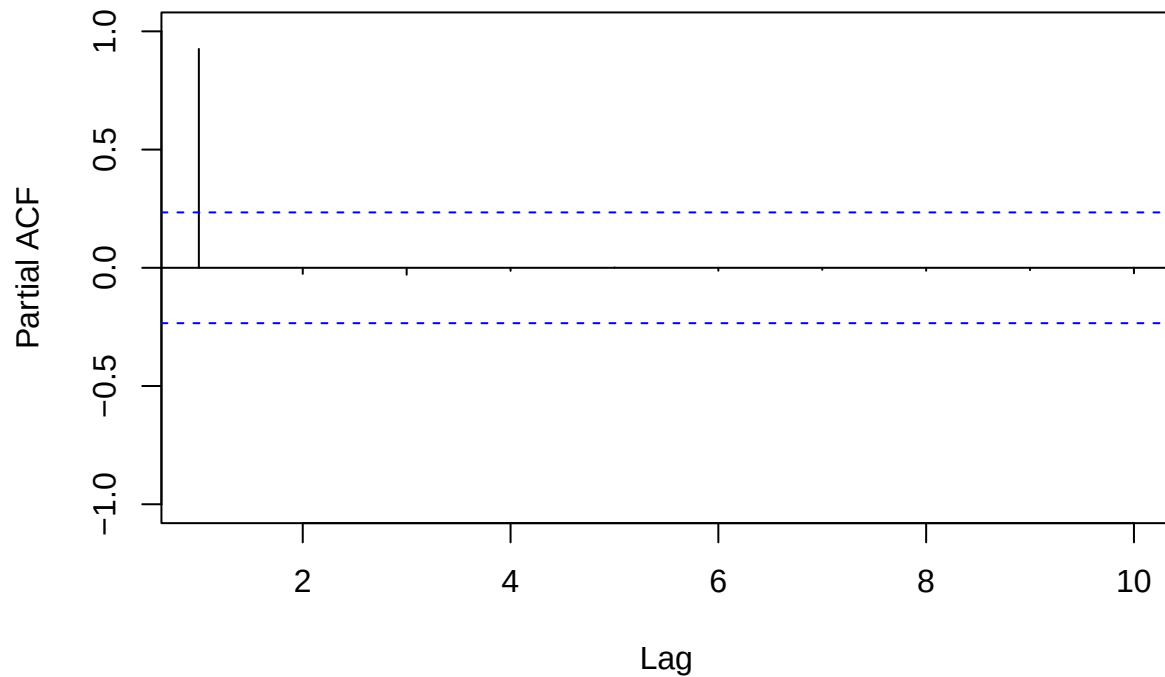
```
# -----  
# (i) Check stationarity using ACF/PACF plot  
# -----  
  
acf(bank_data,  
    main = "ACF of Commercial Bank Real Estate Loan Data",  
    lag.max = 10,  
    ylim = c(-1,1))
```

ACF of Commercial Bank Real Estate Loan Data



```
pacf(bank_data,  
     main = "PACF of Commercial Bank Real Estate Loan Data",  
     lag.max = 10,  
     ylim = c(-1,1))
```

PACF of Commercial Bank Real Estate Loan Data



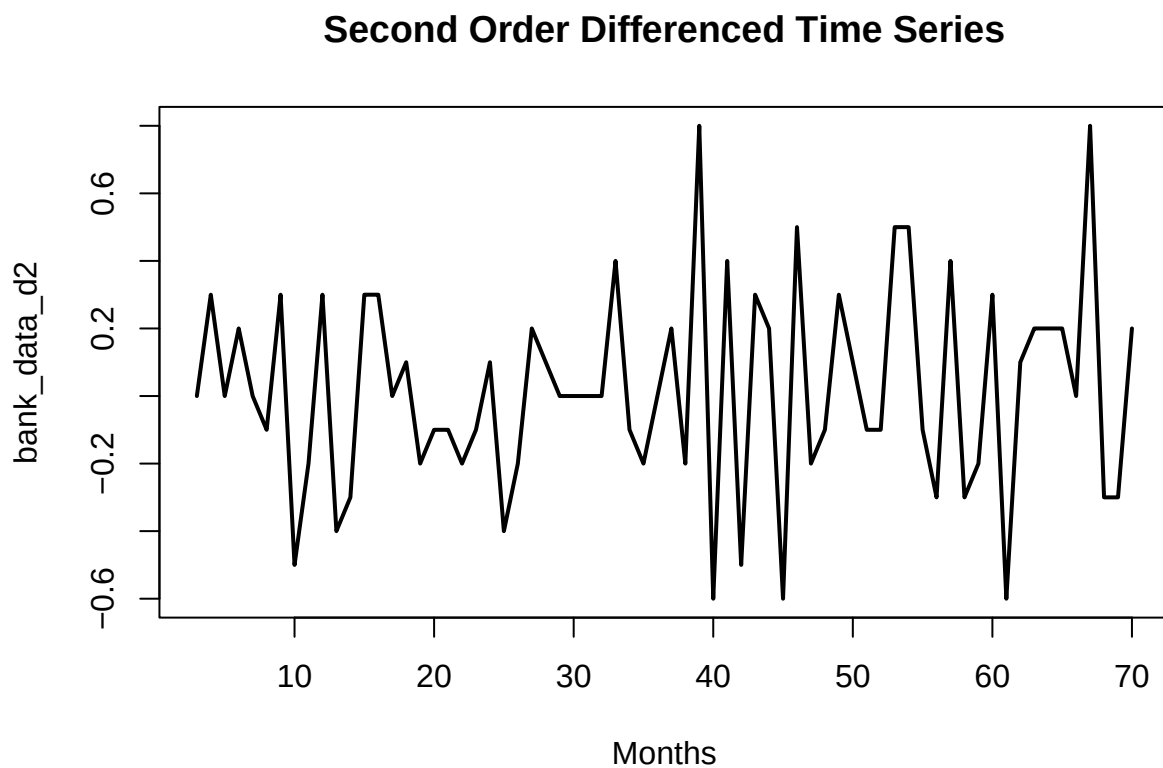
```
# -----  
# (j) Check stationarity using ADF Test  
# -----  
  
adf.test(bank_data)  
  
##  
## Augmented Dickey-Fuller Test  
##  
## data: bank_data  
## Dickey-Fuller = -0.26816, Lag order = 4, p-value = 0.9894  
## alternative hypothesis: stationary  
  
# -----  
# (k) If non-stationary, make it stationary  
# Using difference operator  
# -----  
  
# First differencing  
bank_data_d1 <- diff(bank_data)  
  
adf.test(bank_data_d1)  
  
##  
## Augmented Dickey-Fuller Test
```

```
##
## data: bank_data_d1
## Dickey-Fuller = -1.7533, Lag order = 4, p-value = 0.6755
## alternative hypothesis: stationary
```

```
# Second differencing (since first was not stationary)
```

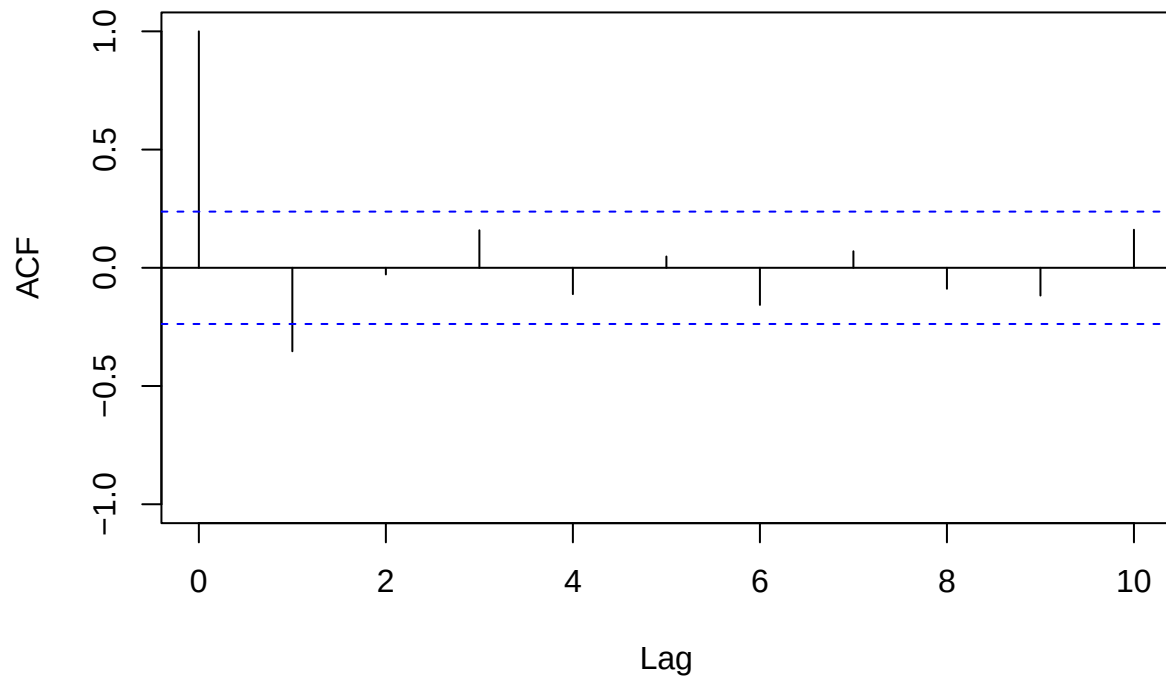
```
bank_data_d2 <- diff(bank_data_d1)
```

```
plot(bank_data_d2,
     main = "Second Order Differenced Time Series",
     xlab = "Months",
     lwd = 2)
```



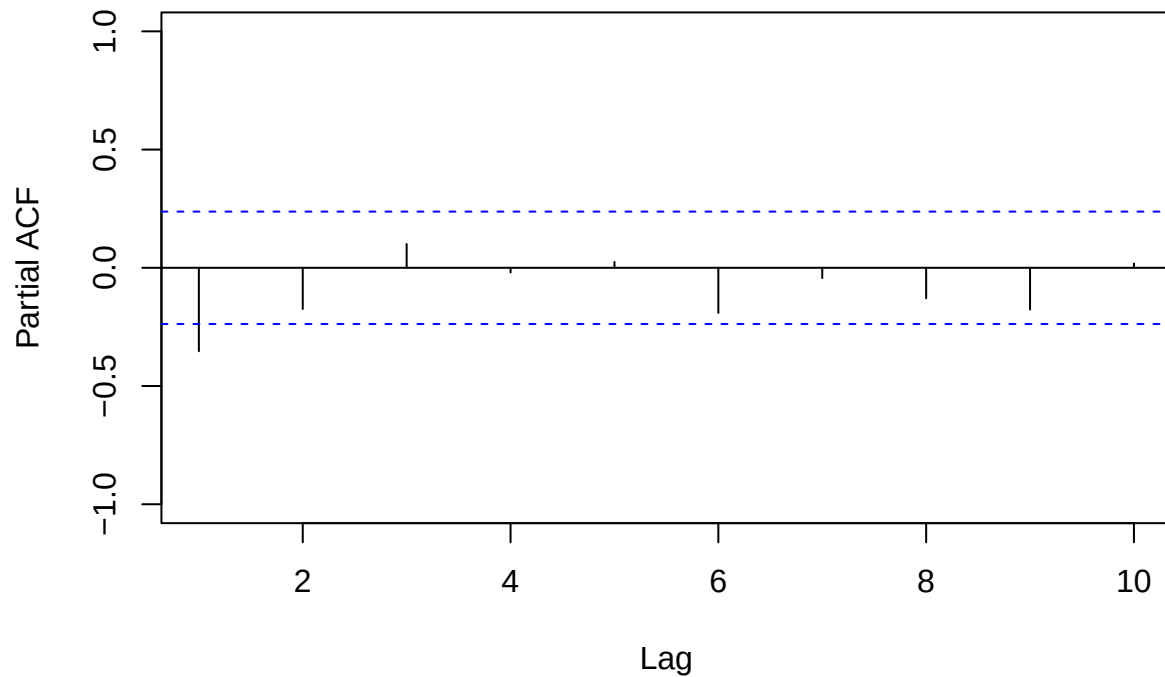
```
acf(bank_data_d2,
    main = "ACF of 2nd Order Difference Data",
    lag.max = 10,
    ylim = c(-1,1))
```

ACF of 2nd Order Difference Data



```
pacf(bank_data_d2,  
     main = "PACF of 2nd Order Difference Data",  
     lag.max = 10,  
     ylim = c(-1,1))
```

PACF of 2nd Order Difference Data



```
adf.test(bank_data_d2)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: bank_data_d2
## Dickey-Fuller = -4.0531, Lag order = 4, p-value = 0.01265
## alternative hypothesis: stationary
```

```
# -----
# (l) Model Identification using ACF/PACF and AIC
# From ACF/PACF of second difference:
# Possible p = 0,1,2
# Possible q = 0,1
# Best model selected: ARIMA(0,2,1)
# -----
```

```
bank_data_fit <- arima(bank_data, order = c(0,2,1))
```

```
bank_data_fit
```

```
##
## Call:
## arima(x = bank_data, order = c(0, 2, 1))
##
```



```
## Coefficients:
##          ma1
##        -0.3723
## s.e.    0.1074
##
## sigma^2 estimated as 0.08089:  log likelihood = -11.07,  aic = 26.13
```

```
# -----
# (m) Fit the data and estimate parameters
# (Already fitted above using arima function)
# -----
```

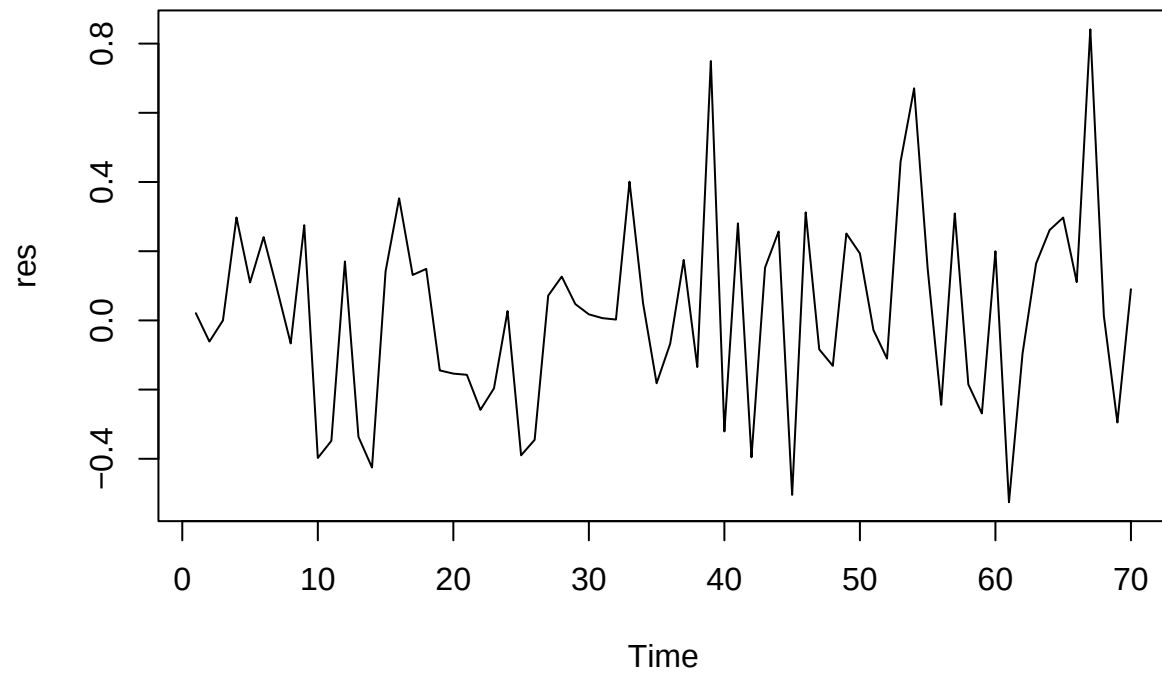
```
# -----
# (n) Check goodness of fit (Diagnostic Checking)
# -----
```

```
# Extract residuals
res <- bank_data_fit$residuals
res
```

```
## Time Series:
## Start = 1
## End = 70
## Frequency = 1
## [1]  0.0207954298 -0.0612681076 -0.0000477981  0.2974844128  0.1097007931
## [6]  0.2407549685  0.0896135857 -0.0666386102  0.2751913095 -0.3975500319
## [11] -0.3480024676  0.1704434072 -0.3365462337 -0.4252915933  0.1416697143
## [16]  0.3527417111  0.1313209497  0.1488890065 -0.1445706444 -0.1538216880
## [21] -0.1572657259 -0.2585478945 -0.1962538707  0.0269373486 -0.3899715909
## [26] -0.3451811284  0.0714937526  0.1266161534  0.0471374748  0.0175486418
## [31]  0.0065331211  0.0024321923  0.4009054722  0.0492516640 -0.1816642742
## [36] -0.0676311427  0.1748218438 -0.1349162012  0.7497725301 -0.3208698671
## [41]  0.2805445051 -0.3955570899  0.1527394662  0.2568628294 -0.5043734562
## [46]  0.3122286104 -0.0837615276 -0.1311832795  0.2511622462  0.1935042941
## [51] -0.0279609786 -0.1104094927  0.4588960450  0.6708407668  0.1497449091
## [56] -0.2442520035  0.3090682954 -0.1849380700 -0.2688499325  0.1999108205
## [61] -0.5255759158 -0.0956647774  0.1643853023  0.2611984161  0.2972406239
## [66]  0.1106586485  0.8411967123  0.0131661146 -0.2950984343  0.0901388596
```

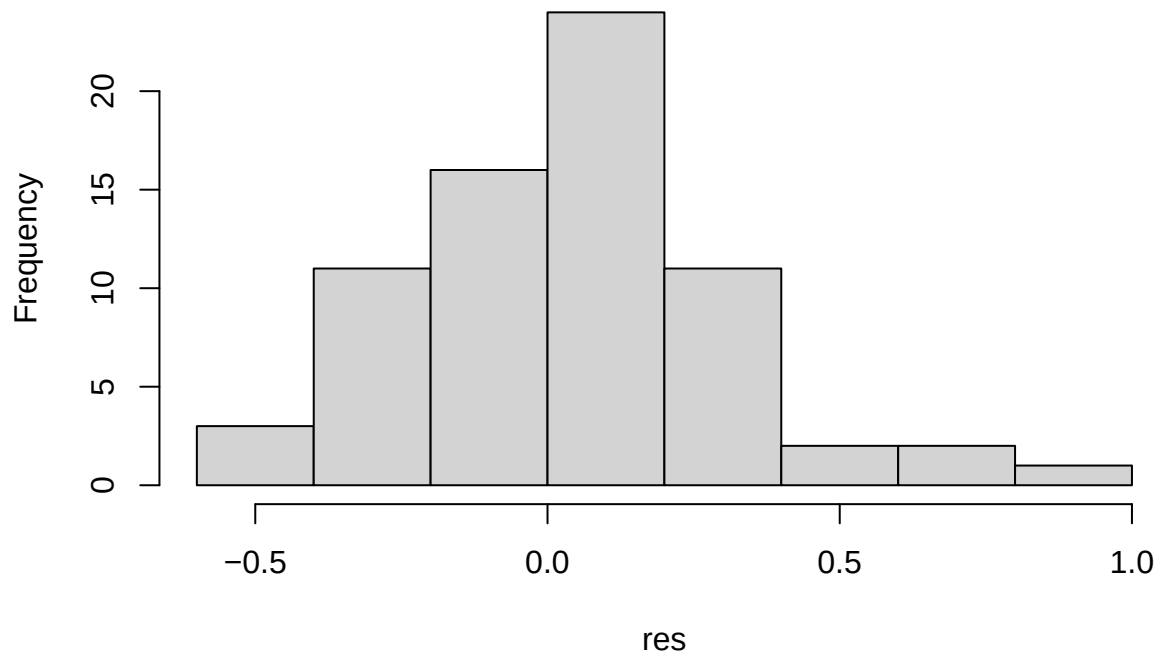
```
# Plot residuals
plot(res,
      main = "Residuals of Commercial Bank Real Estate Loan Data")
```

Residuals of Commercial Bank Real Estate Loan Data



```
# Check Normality of Errors  
hist(res,  
      main = "Histogram of Residuals of Commercial Bank Loan Data")
```

Histogram of Residuals of Commercial Bank Loan Data



```
shapiro.test(res)
```

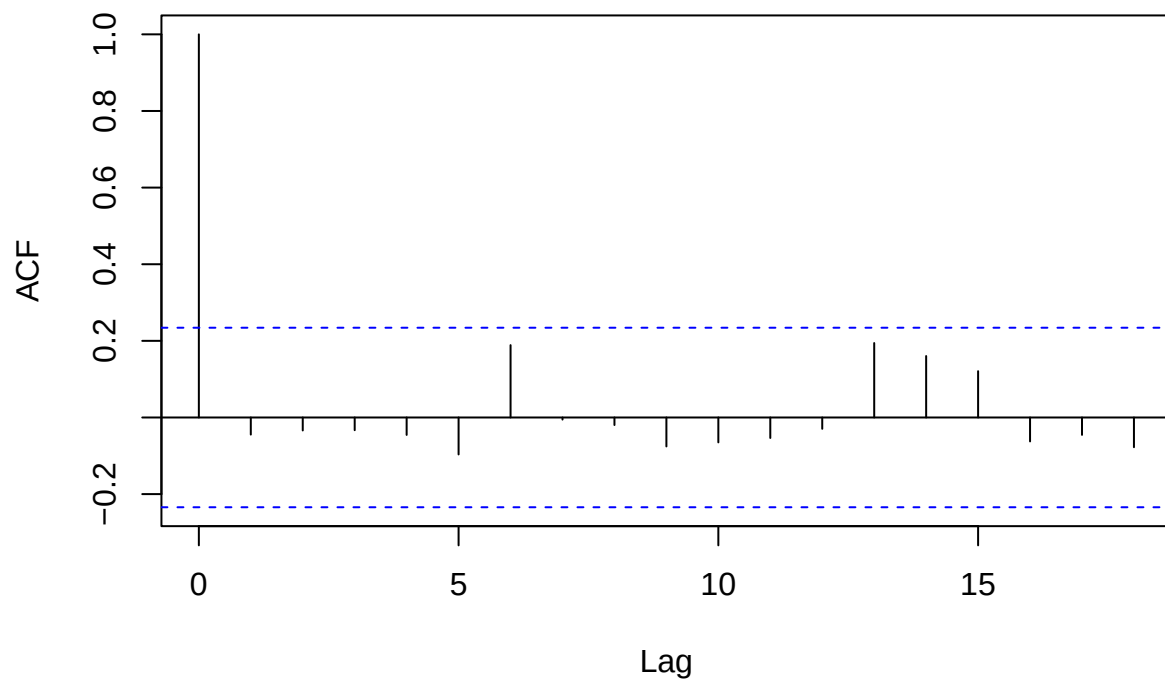
```
##  
## Shapiro-Wilk normality test  
##  
## data:  res  
## W = 0.97676, p-value = 0.218
```

```
# Check Serial Correlation  
Box.test(res,  
  lag = 10,  
  type = "Box-Pierce")
```

```
##  
## Box-Pierce test  
##  
## data:  res  
## X-squared = 8.4829, df = 10, p-value = 0.5818
```

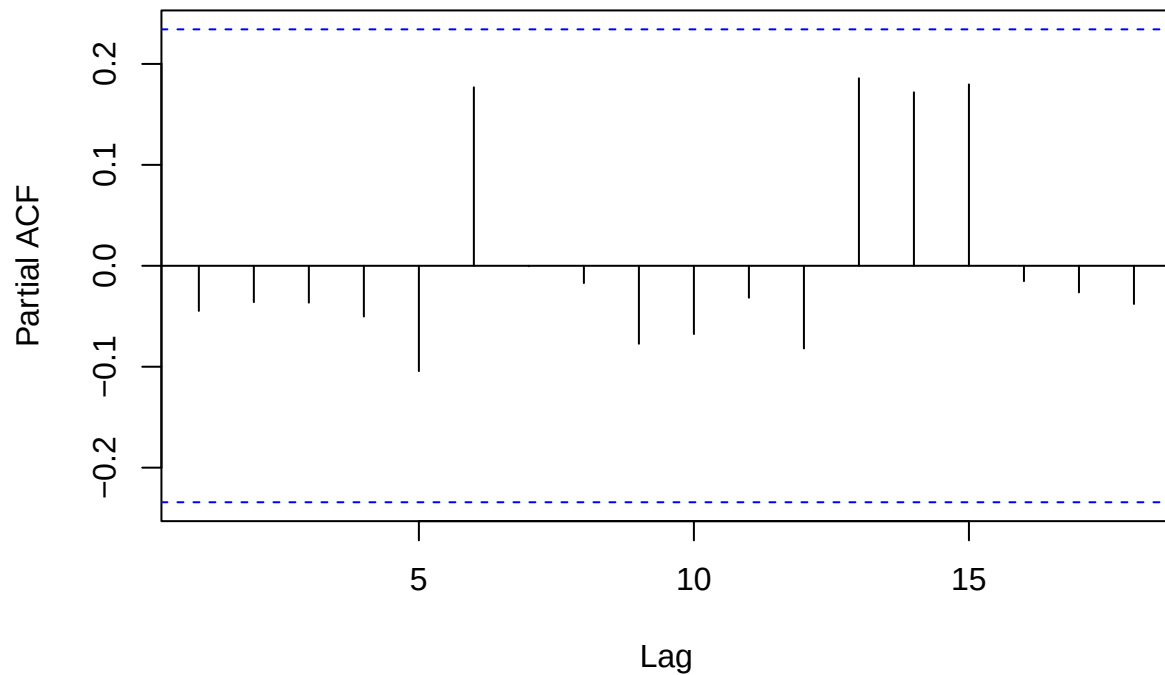
```
# Check Heteroscedasticity  
acf(res^2,  
  main = "ACF of Squared Residuals of Bank Loan Data")
```

ACF of Squared Residuals of Bank Loan Data



```
pacf(res^2,  
      main = "PACF of Squared Residuals of Bank Loan Data")
```

PACF of Squared Residuals of Bank Loan Data



Conclusion

The original data showed a clear upward trend and was non-stationary based on ACF and ADF test.

After first differencing, the series remained non-stationary.

After second differencing, the series became stationary.

Based on ACF/PACF plots of second differenced data, ARIMA(0,2,1) was selected.

Diagnostic checking confirmed:

- Errors are normally distributed
- No serial correlation
- Constant variance

Hence, ARIMA(0,2,1) is a good fit for the data.